AVoiD-DF: Audio-Visual Joint Learning for Detecting Deepfake

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***Abstract*— Recently, deepfakes have raised severe concerns about the authenticity of online media. Prior works for deepfake detection have made many efforts to capture the intra-modal artifacts. However, deepfake videos in real-world scenarios often consist of a combination of audio and visual. In this paper, we propose an Audio-Visual Joint Learning for Detecting Deepfake (AVoiD-DF), which exploits audio-visual inconsistency for multi-modal forgery detection. Specifically, AVoiD-DF begins by embedding temporal-spatial information in Temporal-Spatial Encoder. A Multi-Modal Joint-Decoder is then designed to fuse multi-modal features and jointly learn inherent relationships. Afterward, a Cross-Modal Classifier is devised to detect manipulation with inter-modal and intra-modal disharmony. Since existing datasets for deepfake detection mainly focus on one modality and only cover a few forgery methods, we build a novel benchmark DefakeAVMiT for multi-modal deepfake detection. DefakeAVMiT contains sufficient visuals with corresponding audios, where any one of the modalities may be maliciously modified by multiple deepfake methods. The experimental results on DefakeAVMiT, FakeAVCeleb, and DFDC demonstrate that the AVoiD-DF outperforms many state-of-the-arts in deepfake detection. Our proposed method also yields superior generalization on various forgery techniques.**

***Index Terms*— Deepfake detection, multi-modal, audio-visual, joint learning.**

Manuscript received 7 October 2022; revised 25 February 2023; accepted 7 March 2023. Date of publication 27 March 2023; date of current version 5 April 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3103504; in part by the Shenzhen Science and Technology Program under Grant 20220016; and in part by the National Natural Science Foundation of China under Grant 62261160653, Grant 62122032, and Grant 62172359. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Zhen Lei. *(Corresponding author: Zhihua Xia.)*

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Digital Object Identifier 10.1109/TIFS.2023.3262148

I. Introduction

THE problem of deepfake has received widespread attention in the past few years. Although such techniques have some potential application in some areas such as film making, computer graphics, and virtual reality rending, they can also pose a looming threat to the public. Since deepfake can easily manipulate a person's face and voice, some attackers have employed it to attack authentication systems, impersonate celebrities and politicians, and defraud in finance. Additionally, many freely available deepfake techniques (e.g., Face Swap, Lip-Sync, Text-To-Speech) allow users to easily create forged visuals or audios without any additional expertise. As shown in Figure 1, a forged video of Facebook CEO Mark Zuckerberg has gone viral on Instagram, where the speaker is replaced with the face and voice of Zuckerberg to mislead the audience. Such cases can significantly threaten the counterfeited person as well as the viewers. More importantly, these videos where both visual and audio are maliciously modified could be extremely difficult for detectors to defend. Thus, effective measures should be developed to fight against such multi-modal deepfakes to protect data authenticity.

In recent years, researchers have proposed various deepfake detection techniques to determine whether a visual frame or audio clip is manipulated. Most of these studies mainly focus on uni-modal detection, for instance, examining the intra-modal inconsistency [1], [2], [3], [4], [5], looking for forgery traces based on distortion [3], [6], and observing invisible artifacts of lip's movement [7], [8]. Unfortunately, these approaches are limited to a specific modality and could encounter great difficulties in real-world where visual, audio, and both modalities are forged. There have some other attempts to detect deepfake videos based on multi-modal information. For example, [9] and [10] propose methods to use the unmodified audios as secondary supervisory signals to uncover traces of forgery in visual frames. However, the audio-visual association for deepfake detection is not fully exploited in these previous works. At the same time, many works neglect that both audio and visual can be falsified, which is also an extremely threatening situation. The above-mentioned problems lead to unsatisfying generalization across multi-modalities and various forgery methods.

Deepfake forgery often appears in the multi-modal form in real-world scenarios and poses a greater challenge for deepfake detection. An intuitive idea is to detect multi-modal data

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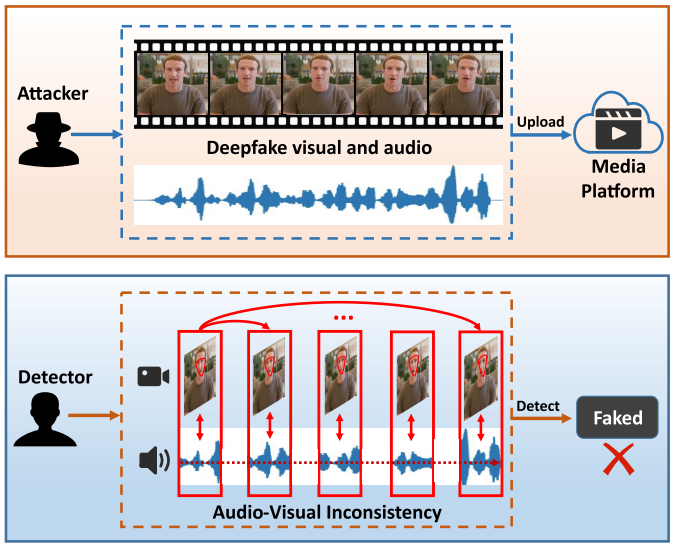


Fig. 1. **The motivation for detecting deepfake with audio-visual.** Attackers aim to generate forged videos with both visual and audio maliciously modified. The uploaded videos can pose a looming threat to everyone on the platform. Our method focuses on exploiting the audio-visual inconsistency among temporal and spatial information for multi-modal deepfake detection. The detector can effectively distinguish whether the media contains forged visual or audio to help against deepfake attacks.

separately and determine accordingly whether the video has been forged or not, e.g., ensemble learning. However, this method is proven to be non-optimal by our experiments. Hard-to-detect deepfake videos often contain carefully constructed audios and corresponding visuals. Uncovering potential correlations and inconsistencies between audio-visuals is critical for detecting these kinds of multi-modal deepfakes. Also, it is common in current deepfake detection tasks, that detectors can not obtain the number of modalities and forgery methods of deepfakes in advance. Therefore, in this opaque situation, it is crucial to better explore the traces of forgery between audio and visual, to help improve the performance of deepfake detectors.

In addition, there are few multi-modal deepfake benchmarks to train multi-modal detectors to solve this practical problem. Exploration on detecting visual deepfakes has produced a number of datasets, while most of them neglect audio deepfakes and multi-modal deepfakes. Although there are some multi-modal deepfake datasets with audio-visual signals, they fail to provide valuable deepfake samples for each modality. In these datasets, the deepfake audio and visual are often unbalanced in terms of number and forgery methods, thus hindering detectors from learning more general multi-modal features. However, few datasets exploit cross-modal techniques for generating forged videos, which are more challenging for deepfake detectors.

Facing these challenges, we first propose a multi-modal dataset DefakeAVMiT for deepfake detection. DefakeAVMiT contains over 6k visuals with corresponding audios, where any one of the modalities may be maliciously modified. Multiple forgery techniques for both audio and visual are adopted to generate the dataset. For example, Faceswap [11], Wave2Lip [12], etc., have been applied to generate forged visuals while TTS [13], Voice Replay [14], etc., are employed to fake audios. The dataset thus provides a more realistic and efficient training environment for multi-modal deepfake detection in real-world. To achieve a more accurate and efficient multi-modal deepfake detection, we propose Audio-Visual Joint Learning for Detecting Deepfake (AVoiD-DF), which incorporates audio-visual signals to capture the inconsistency of deepfakes. AVoiD-DF consists of a Temporal-Spatial Encoder (TSE) for exploring spatial-temporal information of forged visuals or audios, a Multi-Modal Joint-Decoder (MMD) for multi-modal interaction, and a Cross-Modal Classifier for manipulation detection. To comprehensively evaluate the effectiveness of our methods, experiments are conducted on DefakeAVMiT and other two existing multi-modal datasets, i.e., FakeAVCeleb and DFDC. Results have demonstrated that AVoiD-DF achieves great performance on three benchmarks and outperforms other uni-modal and multi-modal approaches. Extensive experiments have also been conducted such as generalization tests and ablation studies. Our main contributions are summarized as follows:

- Multi-modal deepfake detection in real-world scenarios remains a significant challenge. To solve the problem, a benchmark with various deepfake modalities and forgery methods is needed. Thus, we propose the multi-modal dataset DefakeAVMiT, which contains over 6k visuals with corresponding audios, and any one of the modalities has been maliciously modified by multiple forgery techniques. DefakeAVMiT thus provides a more realistic and efficient training environment for multi-modal deepfake detection.
- To better address multi-modal deepfake detection, we design Audio-Visual Joint Learning for Detecting Deepfake (AVoiD-DF). AVoiD-DF focuses on exploring audio-visual inconsistency at temporal and spatial levels to detect multi-modal deepfakes. Experiments demonstrate that the proposed method can obtain a large improvement over the traditional uni-modal system as well as other audio-visual multi-modal methods. Besides, AVoiD-DF achieves promising cross-modal generalization ability.

II. Related Work

A. Deepfake Manipulation and Detection

Deepfakes take their name from the fact that they use deep learning technology to create fake visuals or audios. Recent years have witnessed the rapid development of deepfake techniques, which enable attackers to manipulate videos on a more highly-detailed and multi-modal level. For example, Zhou et al. [15] proposed a deepfake system to generate pose-controllable talking faces with accurate lip synchronization, which was extremely vivid in detail. Similarly, Ji et al. [16] focused on synthesizing high-quality video portraits with emotional dynamics driven by audios. Deng et al. [17] presented an unsupervised approach to retarget the speech of any unknown speaker to an audio-visual stream of known speaker. These powerful video manipulation methods are more diverse, more complex, and more difficult

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to detect and thus present a more challenging problem for deepfake detection.

For image or visual deepfake detection, some efforts focus on special artifacts caused by face forging. For example, Zhao et al. [18] formulated deepfake detection as a fine-grained classification problem and proposed a new multi-attentional deepfake detection network. Since face swapping methods would leave partial context unchanged, Nirkin et al. [19] detected deepfake by comparing two image-derived identities embeddings. Some recent works sought for intra-modal inconsistency by using source images and forged images [1], [20]. However, these methods were not suitable for detecting unseen spoofing techniques. To address this, some approaches aimed to detect deepfake with the help of lip movement and mouth shape, which were more fine-grained features. For example, LipForensics [8] was proposed, which targeted high-level semantic irregularities in mouth movements for detecting generated videos. Yu et al. [21] proposed a commonality learning strategy to learn the common forgery features from different forgery databases. A motion-based fundamental feature extraction network [7] was proposed to extract information about talking habits for authentication. To further improve the transferability, Liu et al. [22] presented a novel Spatial-Phase Shallow Learning method, which combined spatial image and phase spectrum to better capture the up-sampling facial artifacts.

For audio deepfake detection, ASVspoof [14], [23], [24] are a series of competitions that aim to protect automatic speaker verification systems from manipulation. Some methods have been proposed for ASVspoof challenges such as AASIST [25] and wav2vec 2.0 [26]. Based on the LCNN framework, Monteiro et al. [27] proposed a setting for the detection of logical and physical presentation attackers of audio deepfake. Jung and Kim et al. [6], [28] proposed back-end classification models to utilize raw waveforms for speaker verification. Recently, transfer-learning based approach was adopted for unseen audio forgeries [29]. In [2], a capsule network was applied to enhance the generalization of audio spoofing attacks detection. Although [30] proposed a framework for both audio spoof and visual deepfake detection, they still viewed audio and visual as independent tasks and ignored the correlation between audio and visual.

There are also some other works for multi-modal deepfake detection. Wang et al. [31] exploited frequency information as a complementary modality to reveal artifacts that were not perceptible in the RGB domain for deepfake detection. Reference [32] proposed unified audio-visual deepfake detection frameworks based on modality dissonance and dense hierarchical features. Mittal et al. [9] extracted perceived emotion cues between audio and visual modalities for deepfake detection. Cheng et al. [33] addressed the deepfake detection from a voice-face matching view via the intrinsic correlation of facial and audio. Cai et al. [34] proposed a boundary aware temporal forgery detection method for detecting content driven audio-visual manipulations. Agarwal et al. [35] detected the artifacts by comparing the dynamics of the mouth shape with a spoken phoneme. However, this work focused on the explicit representational information between multi-modalities, while ignoring the implicit feature non-synergy. Moreover, these multi-modal methods regarded audios as additional supervision signals and neglected the possibility that audio can also be forged, which is quite common in real-world. More importantly, compared to the explicit audio-visual inconsistency, the implicit potential correlation between multi-modalities is still under-explored and under-utilized.

B. Multi-Modal Representation Learning

Visual and sound are the two most popular modalities for representation learning and there has been a lot of related multi-modal learning work. Recently, audio-visual representations learning has been explored for video captioning, localization, and action recognition, due to the pervasive concurrency property of two modalities. For example, Iashin et al. [36] introduced audio-visual multi-modal to dense video captioning. Wu et al. [37] aimed to localize sound sources in a visual scene and designed a novel *Binaural Audio-Visual Network* (BAVNet). To temporally parse a video into audio or visual event categories, Lin et al. [38] exploited the cross-modality co-occurrence of audio-visual to localize segments of target events. A novel audio-visual transformer [39] framework has been proposed to localize audio-visual events with audio features jointly observed over visual features. For the action recognition task, audio has ideal properties to aid efficient recognition in long untrimmed videos. Based on this, Gao et al. [40] devised a novel framework that focuses on clip-level recognition by distilling from the lighter modalities, a single frame and its accompanying audio. Contrastive learning [41] has also been exploited for action recognition and video understanding by transferring knowledge across heterogeneous modalities between audio-visual. Recently, some weakly-supervised approaches have been explored for multi-modal representation learning with audio-visual such as Audio-visual Transformer (AV-transformer) [39], Audio-Visual Interacting Network (AVIN) [42], and Multi-Modality Self-Distillation (MMSD) [43].

The recent success of audio-visual representations learning proves the inherent and pervasive correlations of multi-modalities, which can be used as extra self-supervision signals. However, due to the heterogeneity of modalities, the combination of audio-visual could also encounter many difficulties. Qian et al. [44] proved that directly transferring the information from one modal to another may lead to modal conflicts and redundancy. Despite the current audio-visual learning systems having obtained huge progress, their performance could degrade dramatically under the more challenging conditions, such as fine-grained retrieval and deepfake detection. We believe that the simple concatenation or fusion operation are not the optimal choice for exploiting information of audio-visual signals. A more inherent relationship and more common inconsistency of two modalities are waiting to be further explored for multi-modal deepfake detection. Inappropriate true-false alignment between multi-modalities could lead to more false positives, thus bringing more noise interference to the detector. Accordingly, detecting deepfake with potential

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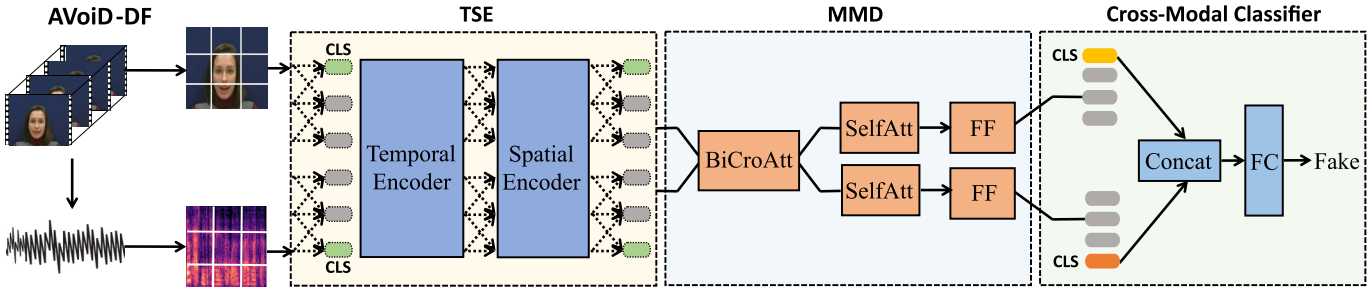


Fig. 2. **Overview of our proposed AVoiD-DF.** In the entry flow, the input visual and audio are first sampled and sliced into patches through data pre-processing. After that, the audio-visual patches are fed into a two-stream Temporal-Spatial Encoder (TSE) to extract features and embed temporal-spatial information. In the fusion flow, Multi-Modal Joint-Decoder (MMD) reads the audio-visual features and fuses them for jointly learning inherent relationships. Finally, the classification tokens of two modalities are selected and concatenated. And the Cross-Modal Classifier together with loss function is exploited to limit the inter-class distance and detect deepfakes.

inconsistencies of audio-visual in the hidden feature space is a more difficult and practical task.

III. The Proposed Method

A. AVoiD-DF Pipeline

Figure 2 gives an overview of our proposed AVoiD-DF. Our method includes three key components, a Temporal-Spatial Encoder (TSE), a Multi-Modal Joint-Decoder (MMD), and a Cross-Modal Classifier. Specifically, the functionalities of each component are as follows:

- The TSE extracts audio-visual features with a two-stream structure. The temporal series and spatial positions are first embedded in synchronized audio and visual. The embedded features are then served as an asset for multi-modal fusion.
- The MMD is adopted for jointly learning audio-visual inconsistency. Based on a Bi-Directional Cross-attention Block, MMD fuses the audio-visual signals and jointly learns the inter-modal disharmony for deepfake detection.
- The Cross-Modal Classifier exploits the output audio-visual tokens of MMD for the final identification of multi-modal deepfakes.

B. Temporal-Spatial Encoder

The trace that distinguishes between real and fake can be unified and concluded as a form of inconsistency, which also exists across frames along the temporal dimension as shown in [5]. Moreover, both visual and audio could be falsified. So we proposed Temporal-Spatial Encoder (TSE) for multi-modal embedding to extract information from temporal-spatial and audio-visual inputs. TSE works in a two-stream manner and consists of two modules: Temporal Encoder E_{tem} and Spatial Encoder E_{spa}. Here, the spatial tokens gather information from spatial patches in a single frame using E_{spa}, and the temporal tokens gather information from patches across frames using E_{tem}. In consideration of computation efficiency and multi-modal input, uniform frame sampling and pre-processing are first conducted on both audio-visual signals. The details of pre-processing are introduced in Section IV-B. After that, each pair of two sampled signals are reshaped from $\mathbf{V} \in \mathbb{R}^{(H\times W\times C)}$ and $\mathbf{A} \in \mathbb{R}^{(H\times W\times C)}$ into the sequence of flattened $\mathbf{V} \in \mathbb{R}^{N\times(P^2\cdot C)}$ and $\mathbf{A} \in \mathbb{R}^{N\times(P^2\cdot C)}$ where C is the number of channels, H and W are the height and width of the input, P^2 is the resolution and N represents the number of total patches. Following transformer [45], the learned classification tokens cls are prepended to both sequences of audio-visual, as shown in Figure 2. The two-stream inputs of TSE can be formulated as:

$$\mathbf{I}^v = [cls^v, \mathbf{V}_{1|1}, \mathbf{V}_{2|1}, \ldots, \mathbf{V}_{t|T}], \tag{1}$$

$$\mathbf{I}^a = [cls^a, \mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_T], \tag{2}$$

where $\mathbf{A}$ and $\mathbf{V}$ denote the two sequences of audio-visual. Following [46], we divide the whole video into T segments and take t key frames in each segment. Here, t denotes the number of frames distilled from the same segment. The final video-level input is obtained by aggregating the t frames of all T clips. cls^v and cls^a are learned classification tokens of visual and audio, respectively.

1) Temporal Encoder E_{tem}: Compared with convolution, the transformer is particularly effective at modeling long-range dependencies and allows the model to attend to all elements in the input sequence. Inspired by this, we develop Temporal Encoder E_{tem} for audio-visual multi-modal embedding along temporal dimensions. E_{tem} composes of a dual-sequence of L transformer layers and each layer comprises layer normalization (LN), Multi-Head Self-Attention (MSA) sub-layer, and Feed Forward (FF) sub-layer, as shown in Figure 3. Particularly, since pre-processed audio is time-sensitive but visual features are time-free, we applied a temporal pooling for the visual modality to maintain the timing consistency of both modalities. The visual features can be formulated as:

$$\mathbf{I}^v = \frac{1}{t}[\mathbf{V}_{1|T} + \mathbf{V}_{2|T} + \cdots + \mathbf{V}_{t|T}], \tag{3}$$

where $\mathbf{V}_{t|T}$ denotes the tth frames distilled from the same visual clips and t is the index of time series. The frames are averaged along temporal dimension and $\mathbf{I}^v$ is aggregate temporal information from multiple frames as training progresses.

To decouple 1D self-attention functions on 2D sequential features, we first operate on each window frame independently. Temporal indexes Tem are added to the input features of audio-visual in MSA to obtain the temporal information, which can be defined as:

$$\mathbf{T}_l^v = \mathrm{MSA}(\mathrm{LN}(\mathbf{T}_{l-1}^v), Tem_i) + \mathbf{T}_{l-1}^v, \tag{4}$$

$$\mathbf{T}_l^a = \mathrm{MSA}(\mathrm{LN}(\mathbf{T}_{l-1}^a), Tem_i) + \mathbf{T}_{l-1}^a, \tag{5}$$

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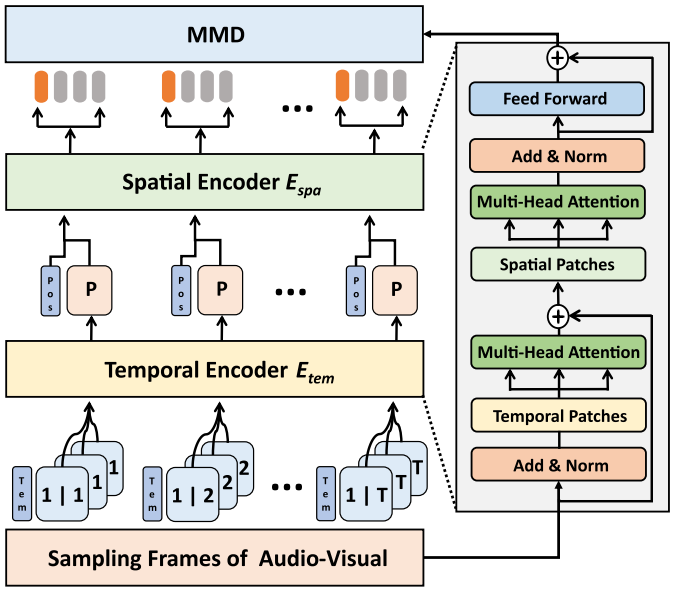


Fig. 3. **The Temporal-Spatial Encoder (TSE).** This module consists of two transformer encoders in series. The first temporal encoder models interactions between time steps of the same window and embeds temporal information. The second spatial encoder produces a latent representation of spatial characteristics per time-index. It thus corresponds to a late fusion of temporal-spatial information. Then the embedded features of two modalities will be output in parallel and sent to the MMD for multi-modal fusion.

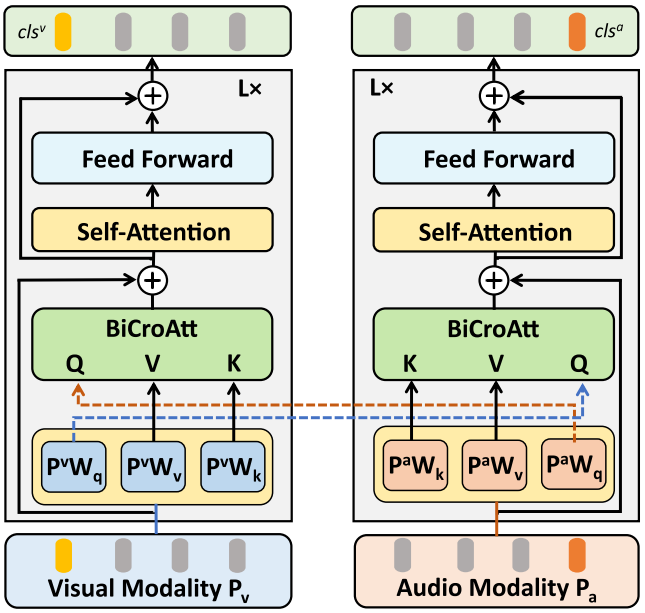


Fig. 4. **The Multi-Modal Joint-Decoder (MMD).** MMD is the modal fusion part of AVoiD-DF. The input embedded patches of vision and sound will be fed through two parallel decoder channels. Each channel has a Bi-directional Cross-attention (BiCroAtt) block followed by self-attention and feed forward layers. BiCroAtt enables the information sharing between two modalities for joint learning.

where $\mathbf{T}^v_{l-1}$ and $\mathbf{T}^a_{l-1}$ are features extracted from the previous encoder layer, l denotes the lth layer. Here, i is the temporal token of audio-visual within the same clips and Tem_i denotes the ith temporal index of audio-visual features. Since audio-visual pairs are extracted synchronously, each pair shares the same temporal index among a batch size. After a linear layer FF, the uni-modal features with temporal information are extracted. Since temporal demension has been added, the output features have the dimension of $\mathbf{T}^v \in \mathbb{R}^{N\times(P^2\cdot C)+1}$ and $\mathbf{T}^a \in \mathbb{R}^{N\times(P^2\cdot C)+1}$.

2) Spatial Encoder E_{spa}: Forgery traces of different methods not only exist in temporal dimensions, but also in spatial channels. Since most deepfake approaches applied the generative adversarial training, there are many obvious traces of forgery in local spatial areas. Based on this, learning the correlations and differences between different regions can better reveal spatial inconsistency. Following ViT [45], we apply the position embedding in Spatial Encoder E_{spa} to obtain the spatial information. E_{spa} aims to embed position tokens to audio-visual features with the input $\mathbf{T}^v$ and $\mathbf{T}^a$ from E_{tem}. Key frames of visual and mel-spectrogram of audio are sliced into patches and position tokens are added.

$$\mathbf{P}^v_l = [\mathbf{P}^v_1, \mathbf{P}^v_2, \ldots, \mathbf{P}^v_n] + Pos_j, \tag{6}$$

$$\mathbf{P}^a_l = [\mathbf{P}^a_1, \mathbf{P}^a_2, \ldots, \mathbf{P}^a_n] + Pos_k, \tag{7}$$

where $\mathbf{P}$ is defined as features of visual and audio. Here, $\mathbf{P}^v_l$ and $\mathbf{P}^a_l$ are spatially embedding features of visual and audio at the lth layer. $\mathbf{P}^v_i$ and $\mathbf{P}^a_i$ represent features of the ith channels at the same position, which share the same position token embedding. Pos_j and Pos_k are learnable position tokens for visual and audio, which are initialized with zeros.

Similar to E_{tem}, we applied pure encoder for encoding spatial information with sef-attention.

$$\mathbf{P}^v = \text{FF}(\text{MSA}(\text{LN}(\mathbf{P}^v_{l-1}))) + \mathbf{P}^v_{l-1}, \tag{8}$$

where $\mathbf{P}^v$ is the visual output of the E_{spa}, which is spatial-wised. LN and FF denote layer-normalization and feed-forward. Same operations for audios are conducted as for visuals. Remarkably, the outputs of the E_{spa} contain both temporal-wised information and spatial-wised information of audio-visual signals and are further exploited for cross-modal joint learning.

C. Multi-Modal Joint-Decoder

In this part, we design the decoder of the transformer [47] to obtain the multi-modal fusion embedding representation and jointly learn the inherent relationships of audio-visual. Although preliminary results have been observed by the simple combination or concatenation of two modalities [44], more delicate multi-modal fusion methods are still not fully explored and compared. Moreover, the distributions of audio and visual are completely different in the feature space. Unsuitable fusion strategies might lead to worse performance on deepfake detection. To address these problems, we introduce MMD, which aims to model multi-modal interaction in a more efficient way. Compared with most of the existing approaches that follow ViT using the pure encoder structure, we apply a joint-decoder with cross-attention mechanism to better jointly learn the potential association of the two modalities collaboratively. MMD constitutes of L layers and each layer has two Self-Attention (SelfAtt) sub-layers, a Bi-Directional Cross-attention (BiCroAtt) block, and two Feed-Forward sub-layers (FF) as shown in Figure 4.

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Concretely, in the BiCroAtt block, two modal feature vectors $\mathbf{P}^v$ and $\mathbf{P}^a$ are given, which are extracted from TSE. Following transformer, we define the queries $\mathbf{Q}$ as $(\mathbf{Q}^v, \mathbf{Q}^a) = (\mathbf{P}^v\mathbf{W}_q, \mathbf{P}^a\mathbf{W}_q)$, the keys $\mathbf{K}$ as $(\mathbf{K}^v, \mathbf{K}^a) = (\mathbf{P}^v\mathbf{W}_k, \mathbf{P}^a\mathbf{W}_k)$, and the values $\mathbf{V}$ as $(\mathbf{V}^v, \mathbf{V}^a) = (\mathbf{P}^v\mathbf{W}_v, \mathbf{P}^a\mathbf{W}_v)$, where $\mathbf{W}$ denotes trainable weights of the BiCroAtt block, $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are trainable factors. To further explore the inconsistency between audio-visual signals, the attention score matrices $\mathbf{Q}^v$ and $\mathbf{Q}^a$ are exchanged between two modalities, which indicate the degree of correlation between audio-visual. The progress of BiCroAtt block can be defined as:

$$\begin{aligned}\mathbf{Z}^a &= \text{BiCroAtt}(\mathbf{Q}^v, \mathbf{K}^a, \mathbf{V}^a) \\ &= \text{softmax}(\frac{\mathbf{P}^v\mathbf{W}_q \cdot (\mathbf{P}^a\mathbf{W}_k)^{\mathrm{T}}}{\sqrt{d_k}})\mathbf{P}^a\mathbf{W}_v, \qquad (9)\\ \mathbf{Z}^v &= \text{BiCroAtt}(\mathbf{Q}^a, \mathbf{K}^v, \mathbf{V}^v) \\ &= \text{softmax}(\frac{\mathbf{P}^a\mathbf{W}_q \cdot (\mathbf{P}^v\mathbf{W}_k)^{\mathrm{T}}}{\sqrt{d_k}})\mathbf{P}^v\mathbf{W}_v, \qquad (10)\end{aligned}$$

where $\sqrt{d_k}$ is a scaling factor and subscripts v and a represent parameters in visual and audio respectively. $\mathbf{Z}^a$ and $\mathbf{Z}^v$ are the outputs of the BiCroAtt block, which denote the fused information of audio-visual. Notably, the output vectors $\mathbf{Z}$ are used as cross-modal features of audio-visual to further obtain the intra-modal forgery traces through SelfAtt sub-layers. The intra-modal local embedding with self-attention can be formulated as:

$$\begin{aligned}\mathbf{Z}^{\hat{a},\hat{v}} &= \text{SelfAtt}(\mathbf{K}, \mathbf{Q}, \mathbf{V}) \\ &= \text{softmax}(\frac{\mathbf{Q} \cdot \mathbf{K}^{\mathrm{T}}}{\sqrt{d_k}})\mathbf{V}, \qquad (11)\end{aligned}$$

where $\mathbf{Z}^{\hat{a},\hat{v}}$ is the intra-modal embedding vector of audio-visual. SelfAtt denotes single-head intra-modal self-attention. Finally, after layer normalization and feed forward sub-layer, the cross-modal audio-visual joint-embedding representation of the lth layer is obtained. The overall of MMD can be formulated as:

$$\mathbf{Z}_l^v = \text{LN}(\mathbf{Z}_{l-1}^v + \text{BiCroAtt}(\mathbf{Z}_{l-1}^v, \mathbf{Z}_{l-1}^a)), \qquad (12)$$

$$\mathbf{Z}_l^{\hat{v}} = \text{LN}(\mathbf{Z}_l^v + \text{SelfAtt}(\mathbf{Z}_l^v)), \qquad (13)$$

$$\mathbf{v}^l = \text{LN}(\mathbf{v}^{l-1} + \text{FF}(\mathbf{Z}_l^{\hat{v}})), \qquad (14)$$

where $\mathbf{Z}_l^v$ and $\mathbf{Z}_l^{\hat{v}}$ denotes the output feature of visual channel from BiCroAtt and SelfAtt, separately. $l \in \{1, 2, \ldots, L\}$ and $\mathbf{v}^l$ is the output of lth layer of joint-decoder. LN denotes the Layer Normalization and FF is short for Feed Forward sub-layer. The operation of audio is the same as above of visual and the learnable classification tokens of two modalities are computed jointly in the process. The final classification tokens of visual and audio are then fused and exploited in Cross-Modal Classifier for detecting manipulation with inter-modal disharmony.

D. Cross-Modal Classifier

In real-world deepfake detection environment, both visual and audio signals can be manipulated, which arouse a higher requirement for network optimization with multi-modal. Moreover, the traces of deepfake not only exist in local features, such as artifacts, but are also present in global features of modalities. We present Cross-Modal Classifier for manipulation detection with the help of global classification token cls. The classification tokens cls^a and cls^v are the general representations of the cross-modal information for audio-visual. Compared with input vectors $\mathbf{V}$ and $\mathbf{A}$, cls provides more global information and excludes background noise. Specifically, we combine the final output of MMD to generate the final classification token for deepfake detection, which can be formulated as:

$$cls^{a,v} = \text{FC}(concat(cls^a, cls^v)), \qquad (15)$$

where concat denotes the concatenation operation and FC is the fully connected layer. $cls^{a,v}$ is the combination of embedded learnable classification token of audio-visual, i.e., cls^a, cls^v. The $cls^{a,v}$ is then further exploited for computing cross-modal loss function.

1) Contrastive Loss: (L_{con}) is set to maximize the similarity of classification tokens corresponding to fake and real labels. It is also exploited to minimize the similarity of classification tokens of samples with the same label using the $cls^{a,v}$ concatenated of two modalities. Still, margin α is introduced to prevent the loss from being dominated by easy negatives. Formally, the contrastive loss over a batch size of N can be denoted as:

$$\begin{aligned}L_{con} = \frac{1}{N^2}\sum_i^N [\sum_{y_i=y_j}^N (1 - \text{Sim}(cls_i, cls_j) \\ + \sum_{y_i \neq y_j}^N max((\text{Sim}(cls_i, cls_j) - \alpha), 0)], \qquad (16)\end{aligned}$$

where cls_i and cls_j are pre-processed with normalization and $\text{Sim}(cls_i, cls_j)$ is the cosine similarity of cls_i and cls_j.

2) Cross-Entropy Loss: (L_{cro}) ensures the learning of fused discriminative features in the same modality by using cls^a and cls^v.

$$L_{cro} = \frac{1}{S^v}\sum_{k=1}^{S^v} l'(cls_k^v, y_k^v) + \frac{1}{S^a}\sum_{k=1}^{S^a} l'(cls_k^a, y_k^a), \qquad (17)$$

where l' represents the cross-entropy loss function, v and a denote the modal types of vision and sound respectively. y is the ground truth of the modality and S denotes the number of tokens in the training set.

3) Additive Angular Margin Loss: (L_{aam}) is used to obtain highly discriminative features for face recognition [48]. It directly optimizes the geodesic distance margin by virtue of the exact correspondence between the angle and arc in the normalized hypersphere. We utilize AAM to optimize the model for deepfake detection as follows:

$$L_{aam} = -\frac{1}{N}\sum_{i=1}^N log\frac{e^{s\cdot cos(\theta_{y_i}+m)}}{e^{s\cdot cos(\theta_{y_i}+m)} + \sum_{j=1, y_j\neq y_i}^N e^{s\cdot(cos(\theta_j))}}, \qquad (18)$$

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TABLE I
THE OVERALL STATISTICS OF DATASETS FOR DEEPFAKE DETECTION. REAL/FAKE DENOTES THE NUMBER OF REAL AND FAKE INSTANCES. DEEPFAKE METHODS ARE COMPUTED WITH ALL MODAL DEEPFAKE METHODS IN THE DATASETS

Dataset	Real/Fake	Deepfake Methods for Visual	Deepfake Methods for Audio	Source
UADFV [3] (2018)	49/49	1	0	YouTube
Deepfake-TIMIT [49] (2018)	0/620	2	0	YouTube
FaceForensics++ [50] (2019)	1,000/4,000	4	0	YouTube
Google DFD [51] (2019)	363/3,068	5	0	Actors
Celeb-DF [52] (2020)	890/5,639	1	0	YouTube
DeeperForensics [53] (2020)	10,000/50,000	1	0	Actors
ASVspoof 2017 [14] (2017)	3,566/14,466	0	1	Website
ASVspoof 2019 LA [54] (2020)	5,370/63,822	0	17	Website
DFDC [55] (2019)	23,654/104,500	7	1	Actors
FakeAVCeleb [56] (2021)	500/19,500	2	2	Actors
DefakeAVMiT (Ours)	540/6,480	5	3	Youtube

where m is the additive margin and N is the batch size. The learned embedding features are thus distributed on a hypersphere with a radius of s. $e^{s \cdot cos(\theta_{y_i}+m)}$ denotes the label score of ground truth and θ_j is the angle between the weight matrix and embedding.

In summary, our model is trained with the sum of Cross-Entropy Loss (L_{cro}), Contrastive Loss (L_{con}), and Additive Angular Margin Loss (L_{aam}) which can be expressed as:

$$L_{all} = \alpha L_{cro} + \beta L_{con} + \zeta L_{aam}\,, \tag{19}$$

where L_{all} is the final cross-modal loss function for manipulation detection. α, β, and ζ are the trainable weight parameters used to balance the loss functions.

IV. DATASETS AND SETTINGS

In this section, we first introduce our proposed novel dataset DefakeAVMiT and existing two multi-modal datasets for deepfake detection in Section IV-A. Then, the details of pre-processing and experiment settings are illustrated in Section IV-B.

A. Datasets

In this subsection, the details of our proposed DefakeAVMiT benchmark are introduced and compared with other existing deepfake datasets.

1) DefakeAVMiT: Exploration on detecting deepfakes has produced a number of detection methods as well as datasets. Existing deepfake datasets mainly focus on uni-modal forgery. For example, Celeb-DF [52] and Google DFD [51] provide thousands of manipulated videos of celebrities as shown in Table I. Some datasets have been proposed to foster research on audio anti-spoofing such as ASVspoof 2019 [54]. However, these datasets have some non-negligible defects and can not satisfy the demands of nowadays deepfake detection for the following reasons. First, with the development of deepfake techniques, both visual and audio can be forged at the same time, which calls for the need to detect multi-modal deepfake. However, most existing datasets have only one modal falsified, while the other modal is real. Second, lip-sync technology can effectively reduce explicit forgery traces, thus boosting the difficulty of deepfake detection. However, few datasets contain synchronized lip-sync video when generating deepfakes. Third, in real-world detection scenarios, deepfake detectors can not know in advance how the target data is forged. Thus, a dataset with more forgery modalities and more forgery methods is expected. Due to the limitations of the quality of previous datasets, we proposed DefakeAVMiT.

We propose a novel benchmark DefakeAVMiT for multi-modal deepfake detection as shown in Figure 5. To generate DefakeAVMiT, we gather real videos from the VidTIMIT [57] dataset, which consists of real visuals and corresponding audios of 43 volunteers, reciting short sentences. Our researchers manually select 43 similar-looking pairs of people with each pair having similar skin color, facial features, and other characteristics. All the selected visuals have clear audio tracks, which are highly synchronized with visual frames. To generate deepfake videos, we exploited the latest forgery and synthetic methods as follows:

- Faceswap [11] generates a 3D template based on computer graphics facial landmarks. This template is blended on a target face and the identity of a person is replaced.
- DeepFaceLab [58] conducts high-quality face-swapping by rigorously training on the large dataset of source and target faces. It consists of two auto-encoders and maps a source face on a target face.
- Wave2Lip [12] is used to lip-sync arbitrary talking face visuals in the wild with arbitrary speech.
- EVP [16] is applied for synthesizing high-quality visual portraits with vivid dynamics driven by audios.
- PC-AVS [15] is applied to generate accurately lip-synced talking faces whose poses are controllable by other visuals. Instead of learning pose motions from audios, PC-AVS leverages another pose source video to compensate for head motions.
- SV2TTS [13] is used for real-time voice cloning, generating speech audio in the voice of different speakers.

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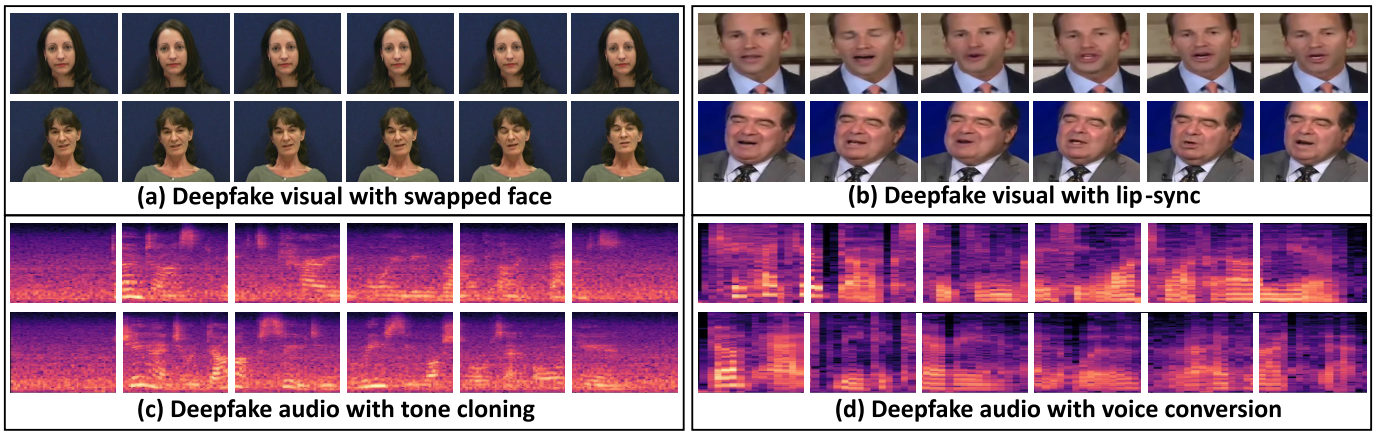


Fig. 5. **The multi-modal deepfakes with both audio and visual being maliciously modified.** With the continuous development of multimedia and short videos, deepfakes often adopt multi-modal forgeries. The multi-modal forgery approaches have more forgery modalities and various forgery combinations, compared with uni-modal ones, making it more difficult to detect.

TABLE II
THE OVERALL ORGANIZATION OF OUR PROPOSED DATASET DEFAKEAVMIT. FIVE VISUAL DEEPFAKES AND THREE AUDIO DEEPFAKES ARE APPLIED FOR GENERATING DEFAKEAVMIT

DefakeAVMiT	Real Audios (A_R)	Fake Audios (A_F)
Real Visuals (V_R)	Raw Data	Tone Cloning (SV2TTS), Voice Replay, AV exemplar autoencoders
Fake Visuals (V_F)	Faceswap, DeepFaceLab, Wave2Lip, EVP, PC-AVS	$V_F \times A_F$

- Voice Replay [14] involves the replay of recordings collected from enrolled speakers to conceal false identity.
- AV exemplar autoencoders [17] employs an unsupervised approach that converts the input speech of any individual into audio-visual streams. It imitates voice, stylistic prosody, and visual appearance of a specific target exemplar speech.

As shown in Table I, the final DefakeAVMiT [1] contains a total number of 6,480 visuals and corresponding audios with 8 different deepfake techniques for visual or audio. The proposed dataset contains the following combined subclasses as shown in Table II: (1) real-visual and real-audio ($V_R A_R$). (2) fake-visual and real-audio ($V_F A_R$). (3) real-visual and fake-audio ($V_R A_F$). (4) fake-visual and fake-audio ($V_F A_F$). When training the models, the combinations of $V_R A_R$ are regarded as positive samples and the remaining three ($V_F A_R$; $V_R A_F$; $V_F A_F$) are considered to be negative. To ensure a high degree of consistency between visual frames and audio tracks, the mean duration of each screen and sentence is cropped as 4.25 seconds or approximately 100 visual frames.

2) FakeAVCeleb: FakeAVCeleb is a recently proposed multi-modal deepfake detection dataset by Khalid et al. [56]. It contains deepfake visuals and respective synthesized lip-synced fake audios. FakeAVCeleb dataset is generated from a selected real YouTube video dataset from VoxCeleb2. FakeAVCeleb used 490 real videos as a base set and generated 19,500 deepfake videos. Faceswap and FSGAN are used to generate swapped deepfake videos. SV2TTS and Wav2Lip are exploited for cloning voices and lip-syncing visuals. However, FakeAVCeleb only contains four different types of audio-visual deepfakes as shown in Tabel I, which cannot provide adequate instances of various deepfake techniques and could hinder the performance of modal generalization.

3) DFDC: Deepfake Detection Challenge (DFDC) [59] dataset is a publicly available face swap video dataset, with over 100,000 total clips sourced from 3,426 actors. DFAE, MM/NN face swap, NTH, StyleGAN, FSGAN, and TTS Skins voice conversion are used to create the deepfake fragments. Unfortunately, there is only one type of deepfake technique for audio forgey in DFDC and the synthesized audios are not lip-synced with the respective visuals.

B. Implementation Details

1) Data Pre-Processing: Pre-processing is conducted separately for visual and audio. For audio, we first compute Mel-Frequency Cepstral Coefficients (MFCC) features to extract speech features using Kaldi toolkit [60]. The MFCC features are stored as three-channel spectrograms to learn the difference between real and fake human speeches. In detail, we apply a 15ms Hann Window and 4ms window shifts, with a sampling rate of 16 kHz. The operations of pre-emphasis and dc offset removal are conducted. Silent fragments are removed using an energy-based voice activity detector. Cepstral filtering is applied for scaling back coefficients to a reasonable range. Followed that, a Sparse Fast Fourier transform (SFFT) with 512 points is conducted to obtain a 2D array of audio's MFCC features per frame.

For visual, we sample visual key frames at intervals of 4ms and obtain 224 × 224 dimension with 3 channels, maintaining the same dimension with audio signals. These visual frames

[1]Our proposed dataset and code are available at https://github.com/SYSU-DISG/AVoiD-DF

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TABLE III
THE OVERALL COMPARISON OF AVOID-DF WITH STATE-OF-THE-ARTS. ACCURACY (ACC) AND AREA UNDER CURVE (AUC) ARE USED TO EVALUATE THE DETECTION PERFORMANCE

Methods	Modality	DefakeAVMiT		FakeAVCeleb		DFDC	
		ACC(%)	AUC(%)	ACC(%)	AUC(%)	ACC(%)	AUC(%)
MesoNet [61] (2018)	visual	80.1	82.5	57.3	60.9	71.7	75.3
Capsule [62] (2019)	visual	68.5	71.4	68.8	70.9	50.2	53.3
Head Pose [3] (2019)	visual	50.6	52.2	45.6	49.2	51.4	55.9
VA-MLP [63] (2019)	visual	56.9	59.8	65.0	67.1	59.3	61.9
Xception [50] (2019)	visual	49.4	51.0	67.9	70.5	46.5	49.9
LipForensics [8] (2021)	visual	73.1	77.2	80.1	82.4	71.3	73.5
DeFakeHop [64] (2021)	visual	83.6	86.1	68.3	71.6	78.9	81.1
CViT [65] (2021)	visual	72.9	74.5	69.7	71.8	62.1	63.7
Multiple-Attention [18] (2021)	visual	85.1	87.3	77.6	79.3	82.5	84.8
SLADD [66] (2022)	visual	73.3	77.5	70.5	72.1	73.6	75.2
LFCC-LCNN [27] (2019)	audio	48.5	50.1	47.4	50.3	44.3	47.8
RawNet [28] (2019)	audio	56.7	59.3	46.8	51.2	54.1	56.2
ECAPA-TDNN [4] (2020)	audio	70.2	72.6	59.8	62.7	67.3	69.8
AASIST [25] (2021)	audio	67.5	69.8	53.4	55.1	64.8	68.4
AVN-J [44] (2021)	audio-visual	84.4	86.2	73.2	77.6	81.1	83.3
MDS [10] (2020)	audio-visual	92.0	94.3	82.8	86.5	89.8	91.6
Emotions Don't Lie [9] (2020)	audio-visual	88.7	91.9	78.1	79.8	80.6	84.4
AVFakeNet [32] (2022)	audio-visual	91.8	93.7	78.4	83.4	82.8	86.2
VFD [33] (2022)	audio-visual	93.4	95.6	81.5	86.1	80.9	85.1
BA-TFD [34] (2022)	audio-visual	92.1	94.9	80.8	84.9	79.1	84.6
AVoiD-DF (Ours)	**audio-visual**	**95.3**	**97.6**	**83.7**	**89.2**	**91.4**	**94.8**

are face-centered and cropped to reduce the impact of the environment on portrait feature extraction. Then, frames of visual and features of audios are paired and processed via TSE, where temporal-spatial information is embedded.

2) Training and Evaluating: All samples from 70% of the paired audio-visuals are randomly selected as the training set and those of the remaining 30% are used as the testing set. Meanwhile, heuristic over-sampling is applied to create the copy of minority samples to balance the positive and negative samples when loading data. As introduced in Section IV-A, we conduct experiments on three benchmarks: DefakeAVMiT, FakeAVCeleb, and DFDC, where audio-visual signals are used as model inputs. Before training, all the datasets are cleaned and filtered by our researchers to ensure the alignment of audio-visual. As mentioned earlier, every pair of audio-visual has two kinds of labels, real or fake. To better simulate real-world detection conditions, we do not label each forgery method explicitly. We train our model for 2000 epochs using an SGD optimizer with a learning rate of 0.01 initially. The varying duration of the videos in some data sets may bring about data imbalance problems. To solve this, we use a downsampling operation for longer video clips to remove the extra part at the end. With 4ms window shifts, each clip contains about 100 pairs of visual and audio.

The operation of the evaluation stage remains the same as the above, with parameters fixed. We report the Accuracy (ACC) and Area Under Curve (AUC) as evaluation metrics to evaluate the detection performance of our model. It should be noted that to avoid the impact of the random seeds on detection performance, all the experiments were repeated for multiple rounds and the average value was recorded.

V. EXPERIMENTS AND RESULTS

A. Comparative Experiments

To evaluate the performance of the proposed method, we compare AVoiD-DF with other approaches for both uni-modal and multi-modal deepfake detection. Experiments are conducted on three benchmarks and results are shown in Table III.

1) Comparison With Uni-Modal Methods for Visual Deepfake Detection: Most early works focus on uni-modal deepfake detection in visual. Tabel III illustrates the comparison of our visual-based model with previous visual-based approaches. On all datasets, our proposed method outperforms MesoNet [61] over 15%, which is a classical deep learning approach for deepfake detection. Recent studies indicate that the combination of capsule networks and dynamic routing algorithm is effective for detecting manipulations, such as

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Capsule [62]. However, the capsule network obtains poor accuracy when encountering unseen deepfake instances. Head Pose [3] is proposed to reveal errors when 3D head poses are estimated from the face images, using an SVM classifier. We notice that Head Pose gets a poor AUC score at about 50% on three benchmarks, due to its lack of generalization. LipForensics [8] targets high-level semantic irregularities in mouth movements to detect forgery traces. However, it encounters great difficulties since some lip-sync methods have been applied in three datasets, which have high-precision indistinguishable lip movement simulations. VA-MLP [63] is a simple classifier for detecting visual artifacts with the help of eye color, missing details in the eye and teeth areas, face border and nose tip, etc. On all benchmarks, we convincingly beat the VA-MLP by around 20%, with the help of a more general and effective supervision signal. Xception [50] uses additional domain-specific knowledge to improve forgery detection. It can be observed that our method boosts the accuracy from 46.5% to 91.4% in contrast with Xception on DFDC.

Recently, some convincing works have been conducted for building generalizable deepfake detectors and some of them have achieved preliminary promising results. For example, Multiple-Attention [18] regards deepfake detection as a fine-grained classification problem and proposes a multi-attentional deepfake detection network. We can observe that our proposed method outperforms the previous detector Multiple-Attention by an average of 8.4% on three datasets in terms of ACC. DeFakeHop [64] introduces successive subspace learning to deepfake detection. Our method beats it by 11.7% in terms of AUC on DefakeAVMiT. CViT [65] exploits a convolutional vision transformer for the detection of deepfakes, which shares a similar backbone to ours. Remarkably, AVoiD-DF outperforms CViT on three benchmarks under all metrics. Moreover, self-supervised learning has also been used for deepfake detection such as SLADD [66], which achieves good performance on unseen methods. Our method applies an audio-visual cross-supervised strategy for detecting forgery and beats SLADD on three datasets under all metrics.

2) Comparison With Uni-Modal Methods for Audio Deepfake Detection: ASVspoof aims to protect automatic speaker verification systems from manipulation. Some early works have been proposed to resist deepfake audio attacks. LFCC-LCNN [27] introduces a convolutional neural network architecture for the detection of attacks to voice biometrics systems. Our method outperforms LFCC-LCNN on three benchmarks. Unlike other methods, RawNet [28] uses raw waveforms for text-independent speaker verification with a simple, effective back-end DNN classifier. AVoiD-DF obtains 38.6% higher accuracy than RawNet on DefakeAVMiT. Lately, some more effective networks have been introduced for audio deepfake detection. ECAPA-TDNN [4] is a novel model for speaker verification with emphasized channel attention, propagation, and aggregation in TDNN. Remarkably, our proposed model beats the method by an average of 25.5% in terms of AUC on three benchmarks. AASIST [25] is proposed to model artifacts spanning temporal and spectral domains, but gains poor accuracy at an average of 61.9% on three benchmarks. Since audios recorded within real-world videos have tremendous noise, the audio signals in datasets could be intermittent and indistinct, which could hamper the performance of speaker verification.

3) Comparison With Multi-Modal Methods for Deepfake Detection: These days some multi-modal methods have been proposed for deepfake detection, incorporating both audio-visual signals. For example, Emotions Don't Lie [9] extracts and analyzes the similarity between the two audio and visual modalities. However, given different representations of expressing perceived emotions, Emotions Don't Lie also finds a mismatch in the modalities of real videos. Our proposed AVoiD-DF outperforms Emotions Don't Lie at an average of 8.5% in terms of AUC on three datasets. MDS [10] is another proposed method for deepfake detection based on the dissimilarity between the audio and visual modalities. By adopting the dissonance rationale for deepfake detection, the audio-visual dissimilarity is computed to perform a fine-grained analysis. Our method obtains a 3.3% higher accuracy on DefakeAVMiT than MDS. A great problem is that all the above methods exploit audio as an extra supervision signal but ignore that audio can also be forged. Furthermore, deepfake methods for audios could be various and even harder to detect than visual ones. AVN-J [44] is a multi-modal learning strategy for person verification. However, the spatial-temporal information has not been considered in AVN-J, which leads to the degeneration in deepfake detection as shown in Tabl III. It also can be observed that our proposed method outperforms very recent works (i.e., AVFakeNet [32], VFD [33], and BA-TFD [34]) on multi-modal deepfake detection with more effective model design. To this end, compared with all the existing multi-modal methods, our proposed AVoiD-DF obtains state-of-the-art performances.

B. Generalization Tests

The generalization ability is an important indicator of the superiority of a deepfake detection approach. Recent studies have observed that current detection methods are still insufficient to be applied in real-world scenes. Since most of the proposed deepfake detectors are designed to detect specific deepfakes, they encounter a great obstacle when facing unseen forgery methods, e.g., Head Pose [3] and LipForensics [8]. These methods focus on capturing the specific artifacts of known deepfake techniques, which is not effective in real-world scenarios. Thus the generalization of deepfake detectors should be further discussed and tested.

To validate the experimental performance of our proposed model on the unseen deepfakes, we conduct a cross-dataset evaluation on three benchmarks and compare the results with other deepfake detection approaches. Notably, DefakeAVMiT has 8 kinds of deepfakes for both audio-visual, FakeAVCeleb has 4, and DFDC has 8. The deepfake methods for each of these datasets are partially different, e.g., DFDC gets 1 audio forgery and DefakeAVMiT has 3. Thus, the validation dataset has unseen deepfakes that the model has not seen before, which raises higher demands on the generalization of the model.

Table IV presents the cross-dataset evaluation results on three benchmarks, where we exploit each of the benchmarks

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TABLE IV
THE AUCS OF CROSS-DATASET EXPERIMENTS. THE TRAINING SETS AND TESTING SETS FOR CROSS-DATASET DEEPFAKE DETECTION ARE SHOWN IN ROW 1 AND ROW 2, RESPECTIVELY

Methods	DefakeAVMiT		FakeAVCeleb		DFDC	
	DFDC	FakeAVCeleb	DefakeAVMiT	DFDC	DefakeAVMiT	FakeAVCeleb
LipForensics [8]	58.8	54.4	52.5	53.1	51.6	49.2
CViT [65]	61.0	59.3	56.5	56.7	47.2	45.5
MesoNet [61]	60.4	62.5	42.9	41.2	57.7	54.1
MDS [10]	82.2	80.5	76.3	73.8	76.5	72.9
AVoiD-DF (Ours)	**90.3**	**88.7**	**83.1**	**80.6**	**84.4**	**82.8**

TABLE V
OVERALL COMPARISON OF OUR PROPOSED MMD WITH OTHER MULTI-MODAL FUSION METHODS

Fusion Methods	Fusion Level	DefakeAVMiT		FakeAVCeleb	
		ACC	AUC	ACC	AUC
Ensemble	score-level	70.6	72.4	59.7	61.1
Contrastive	score-level	71.9	74.6	61.6	65.3
PC	pixel-level	66.0	68.1	56.4	58.9
MFB [67]	feature-level	72.1	75.8	65.3	67.8
AVN-F [44]	feature-level	72.5	73.2	63.0	66.3
AVN-E [44]	embedding-level	79.8	83.6	72.1	74.5
Ours	**embedding-level**	**95.3**	**97.6**	**83.7**	**89.2**

for training, and the other two datasets for testing. It can be observed that our model outperforms all the other comparative methods on DefakeAVMiT, achieving 90.3% and 88.7% in terms of AUC on two different evaluation datasets. When training on FakeAVCeleb and DFDC, AVoiD-DF also obtains the optimal results with an average value of 81.8% and 83.6%. Compared with other approaches, our model also achieves the best generalization. For example, MesoNet [61] and CViT [65] get over 20% and 15% decline when evaluating on unseen deepfakes, respectively. Approaches that focus on local areas, e.g., lips movement, head pose, eyes, and emotions, also have an obvious decline such as LipForensics [8]. It is possible that test data is lowly correlated in those areas with training data since datasets are very likely to be collected from different sources. MDS [10], the multi-modal deepfake detection method, obtains a better result than other uni-modal methods. Still, our proposed AVoiD-DF outperforms the MDS on cross-dataset evaluation. We can observe that AVoiD-DF significantly improves the generalization of the model and achieves great results on unseen deepfakes.

C. Ablation Studies

In this subsection, we discuss the effects of each component in AVoiD-DF. Experiment results are shown in Table V and Table VI. The multi-modal fusion strategy for deepfake detection has also been discussed to comprehensively investigate the detection performance of the proposed MMD.

1) Effectiveness of Audio-Visual Joint Learning: Although some previous works have exploited audio as extra supervision for visual deepfake detection, a more realistic situation has been neglected, where audio, visual, or both are forged. In this paper, our proposed method reveals the inherent relationships between audios and visuals, incorporating both modalities for deepfake detection. For conducting uni-modal ablation studies, we first modify the inputs of the model to two sets of the same modalities. As shown in Table VI, uni-modal learning of visual or audio gains poor performances at an average of 75.2% and 60.8% in terms of AUC on three benchmarks. In a contrast, our proposed multi-modal model achieves promising results, which demonstrates a more powerful and robust deepfake detection system with the help of audio-visual. Visualizations of comparative experiments of uni-modal and multi-modal are shown in Figure 6.

2) Effectiveness of MMD: Although good performances are observed by simply combining audio-visual information as illustrated in AVJ [44], more precise and robust fusion strategies still need to be investigated for deepfake detection. To examine the effectiveness of the proposed MMD strategy, six variants of multi-modal fusion strategies have been analyzed, including 1) Ensemble learning, which trains two modal deepfake detectors separately and votes on the decision layer; 2) Contrastive learning, which is used by MDS, aims to detect deepfakes by comparing disharmony between the two modalities; 3) PC, denotes Pixel Concatenation, which is a simple fusion strategy to concatenate audio-visual signals at pixel-level. Since PC has not been implemented on deepfake detection before, we modified our model to have it trained using audio-visual modalities; 4) MFB, which denotes multi-modal factorized bilinear pooling, is exploited to combine multi-modal features at high-dimensional representations; 5) AVN-F [44], a feature level audio-visual deep neural network for person verification and 6) AVN-E [44], an embedding level audio-visual deep neural network for person verification.

Table V presents the ablation results of multi-modal fusion methods on two benchmarks. We can observe that PC obtains the lowest results in terms of ACC and AUC, due to the heterogeneity of audio-visual modalities. The simple concatenation could hinder the performance of detectors since multi-modalities have different distributions in feature spaces. Moreover, ensemble learning and contrastive learning are two

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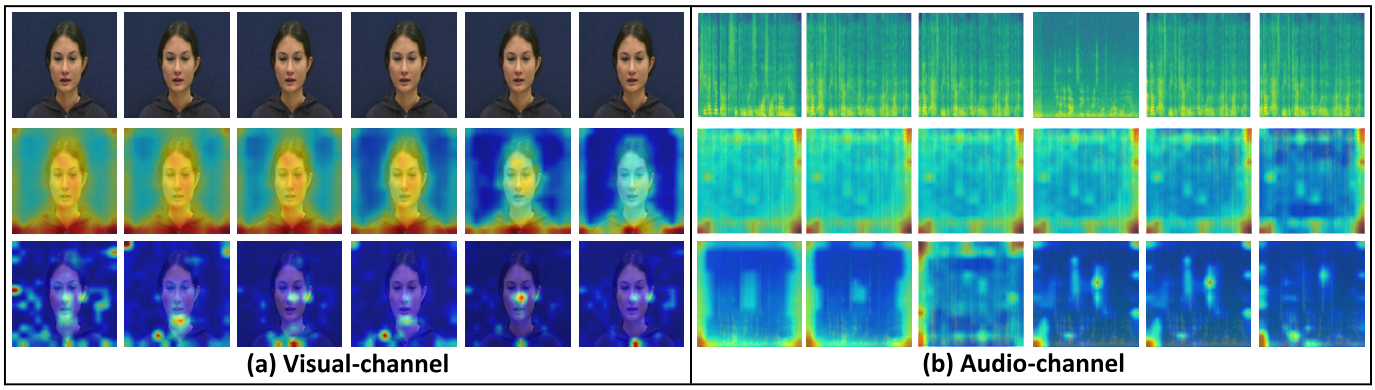


Fig. 6. **The visualization result of uni-modal learning and multi-modal joint learning.** Row 1 denotes the raw data inputs, which are sampled from our proposed dataset. Row 2 represents the distribution of attention values with uni-modal input. Apparently, deepfake forgeries are not well detected due to the double interference from background noise and non-forged areas. Row 3 is the distribution of attention values with the fusion of two modalities in our model. We can conclude that the audio-visual multi-modal help with locating forged area through a more refined local and global attention.

TABLE VI
ABLATION STUDIES FOR THE EFFECTIVENESS OF EACH MODULE IN OUR PROPOSED METHOD AVOID-DF ON THREE BENCHMARKS

Ablation Model	DefakeAVMiT		FakeAVCeleb		DFDC	
	ACC(%)	AUC(%)	ACC(%)	AUC(%)	ACC(%)	AUC(%)
visual only	75.6	77.5	70.3	72.4	74.2	75.8
audio only	62.5	64.3	55.8	57.2	57.4	60.9
w/o TSE	80.2	82.4	70.3	73.7	78.1	82.2
w/o MMD	76.8	79.2	68.5	70.2	74.6	77.8
w/o Classifier	85.8	87.6	74.1	79.4	82.5	86.0
w/o L_{cro}	94.6	96.9	83.1	88.5	90.2	94.1
w/o L_{con}	92.7	94.2	81.4	86.3	89.8	92.1
w/o L_{aam}	93.1	95.2	82.0	87.6	90.5	93.9
AVoiD-DF (Ours)	**95.3**	**97.6**	**83.7**	**89.2**	**91.4**	**95.8**

methods that fuse multi-modal at score-level and both of them get poor performance on two benchmarks. It is inferred that the co-subtle relationship and information of audio-visual have not been fully exploited, which leads to bad results. Compared with feature-level fusion methods MFB and AVN-F, our proposed MMD gains a higher accuracy by 23.2% and 22.8% on two datasets. Remarkably, our method outperforms AVN-E on FakeAVCeleb by 14.7% in AUC, which fuses audio-visual signals at the embedding-level. We believe that it is the designed MMD that boosts the results, which fully utilizes potential information of audio-visual.

3) Effectiveness of TSE: Recently, some works have exploited spatial-temporal inconsistency for a more general deepfake detection such as STIL [5]. The trace that distinguishes between real and fake can be existing not only within each frame but also across frames along the temporal dimension. Thus, we propose TSE for obtaining temporal-spatial inconsistency between audio-visual signals. As shown in Table VI, TSE plays an important role in deepfake detection and significantly improves the accuracy from 70.3% to 83.7% on FakeAVCeleb. We attribute this achievement to the two-stage spatial-temporal embedding, which explores the temporal-spatial inconsistency at the frame-level.

TABLE VII
ABLATION STUDIES FOR THE THREE TRAINABLE WEIGHT PARAMETERS OF THE LOSS FUNCTIONS

Settings	Parameters			DefakeAVMiT	
	$\alpha(L_{cro})$	$\beta(L_{con})$	$\zeta(L_{aam})$	ACC(%)	AUC(%)
fixed	0.3	0.3	0.3	95.0	97.1
	0.2	0.3	0.5	94.8	97.0
	0.3	0.2	0.5	94.1	95.9
	0.5	0.2	0.3	92.9	94.5
	0.5	0.3	0.2	93.3	95.4
	0.2	0.5	0.3	95.2	97.3
	0.3	0.5	0.2	95.0	97.1
trainable	0.2	0.4	0.4	95.3	97.6

4) Effectiveness of Cross-Modal Classifier and Loss Functions: Cross-Modal Classifier together with loss function is designed for final detection, combining audio-visual multi-modalities. The Classifier aims to judge deepfakes with the help of classification tokens from both modalities. Experimental results show the effectiveness of the Classifier, which helps

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with reducing modal redundancy and excluding noise. Further, the bad performances of row 5, 6 and 7 in Table VI indicate the importance of the cross-modal loss function, which takes the weighted sum of Cross-Entropy Loss (L_{cro}), Contrastive Loss (L_{con}) and Additive Angular Margin Loss (L_{aam}). More specifically, to better explore the relationship and importance among three loss functions, we perform a quantitative analysis of the three trainable parameters, as shown in Table VII. We set the three weight parameters as fixed and trainable, respectively. Experimental results show the importance of each loss function and the optimal performance is achieved with the trainable parameter setting. The combination of Cross-Modal Classifier and loss functions improves the performances of our method.

VI. Conclusion

In this paper, we propose Audio-Visual Joint Learning for Detecting Deepfake (AVoiD-DF), which jointly learns audio-visual inconsistency for multi-modal deepfake detection. We introduce Temporal-Spatial Encoder (TSE) for feature embedding, which aims to explore the inconsistency of audio-visual at temporal-spatial levels for deepfake detection. The Multi-Modal Joint-Decoder (MMD) is presented for jointly learning multi-modal interaction and fusing audio-visual information. Further, we propose Cross-Modal Classifier for manipulation detection and the cross-modal loss function for judging deepfakes. A novel benchmark DefakeAVMiT is provided for deepfake detection, hoping it can bring about benefits for further research in this community. Experimental results show that our method achieves superior performance and outperforms other deepfake detection methods. It is proved that the prevalent inconsistency in multi-modalities can be an effective cue for deepfake detection in real-world scenarios.

The multi-modal deepfake detection remains an open problem and a challenging task. How to uncover inter-modal inconsistency more efficiently is the key to detect multi-modal deepfakes and improve model generalization. The proposed AVoiD-DF can be an effective solution. In future work, we will commit to forming a universal approach and enhancing the interpretability of the model.

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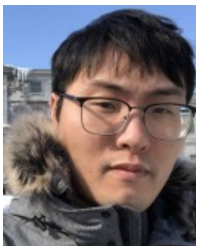


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